When Does SHAP Outperform LIME? Model- and Configuration-Dependent Results from a Paired Tabular Benchmark

Abstract

Choosing a post-hoc explainer for a tabular model trades faithfulness, stability and computational cost, yet comparisons under matched conditions are scarce. This paper reports a paired benchmark of SHAP, LIME, Anchors and DiCE in which every explainer sees the same model, test instances and seed on one recorded host: five model families on the Adult dataset (confirmatory), six further datasets and a configuration sweep (exploratory). Cost is measured in time and in model evaluations, which do not depend on the hardware. On Adult, SHAP's fidelity exceeds that of each alternative in all 25 model-seed cells (d_z=3.3 or more), but its stability advantage over LIME and Anchors is not significant; DiCE ranks last on every quality endpoint. On the six further datasets SHAP keeps the highest fidelity, while LIME is the more stable explainer, as it is on Adult with category-preserving noise or a wider kernel; SHAP's fidelity lead holds over every configuration tested. KernelSHAP takes 3 to 25 times LIME's wall time, whereas TreeSHAP's cost follows tree depth: faster than LIME on XGBoost, and on the deeper random forest faster or slower by dataset. The choice of explainer therefore depends on the model, the configuration and the latency budget.

1 Introduction

A team that deploys a tabular classifier and must explain its predictions faces a concrete decision: which post-hoc explainer to use, for which model, under which latency budget. The candidates differ in the kind of explanation they return. LIME [1] and SHAP [2] return feature attributions, Anchors [24] returns a rule and DiCE [25] a counterfactual. They also differ in how faithfully the explanation tracks the model, how stable it is under small changes of the input, and how much computation it needs [40, 37, 42]. Theory does not settle the choice across model families and configurations [3, 6], and the empirical guidance is weaker than the popularity of the methods suggests: evaluations use different metric families, and the ranking of ex-

plainers often depends as much on the metric and the configuration as on the methods [4, 5, 7].

Two gaps follow. The first is *control*. Comparisons in which all explainers see the same model, the same instances, the same seed and the same host, so that a difference can be attributed to the explainer, are scarce: published SHAP–LIME comparisons are often case studies in a single domain [34, 35]. Cost needs the same control: it transfers to another deployment only when it is measured on a stated host, as item 2c of the REFORMS checklist for machine-learning-based science recommends [43], and when it is also counted in a unit that does not depend on that host, such as the number of model calls. The second gap is *configuration*. Results are usually reported for one configuration, although the hyperparameters that build LIME's neighborhood [44] or bound KernelSHAP's sampling [42] can change the outcome. Without control and without more than one configuration, a recommendation cannot state the conditions under which it holds.

This paper addresses the two gaps with three contributions. The first closes the gap in control, the second the gap in configuration, and the third states what the results support once both are closed:

1. **A controlled, paired benchmark of four explainers** (SHAP, LIME, Anchors and DiCE) on five model families, with every run on one recorded host, and cost reported as time on that host and as the number of model evaluations, a measure that does not depend on the hardware (Sections 4.1 and 4.2).
2. **Configuration-aware results**: a sweep over the LIME kernel width and sample size and over the KernelSHAP budget and background, and controls that test whether the stability measurement itself is valid (Sections 4.3 and 4.4).
3. **Bounded recommendations**: every claim is labelled as confirmatory or exploratory and stated with the models, datasets and configurations under which it holds (Section 5).

The paper makes ranking claims under a stated protocol. It does not claim that an explanation is correct for a task, and it does not test whether people decide better with any explainer.

2 Background and Related Work

LIME fits an interpretable surrogate g to the model f on perturbed samples around an instance x, weighted by a proximity kernel π_x, as in Eq. (1):

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g), \tag{1}$$

where G is the class of surrogates (sparse linear models here), $\mathcal{L}$ measures how poorly g reproduces f on the weighted samples and $\Omega(g)$ penalizes surrogate complexity [1]. The kernel width, the number of samples and

the feature budget shape the neighborhood, and the sampling makes the explanation stochastic. SHAP assigns additive attributions $f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i$ that satisfy local accuracy, missingness and consistency, where ϕ_0 is the expected output over a background sample and M the number of features [2]. KernelSHAP estimates them by a weighted regression over sampled feature coalitions and applies to any model; TreeSHAP computes them from the structure of tree ensembles. Anchors searches for a rule that fixes the prediction with a target precision [24]; DiCE searches for a counterfactual instance with the opposite prediction [25].

Quantitative evaluation relies on proxy metrics: faithfulness under feature removal [10, 11], robustness to small input changes [12] and sparsity, standardized in toolkits and benchmarks [13, 14]. Removal-based scores depend on the removal scheme [8], attributions can pass perturbation tests and fail sanity checks [15], LIME and SHAP can be manipulated [46], and Shapley attributions are hard to read as importance when features are dependent [47]; feature attribution has provable limits as a basis for inferring model behavior [48].

Recent work compares SHAP and LIME in applied settings, among them education [34], software engineering [35] and mixed-type tabular models [36], and across tabular and image classifiers [37]; their respective strengths are reviewed in [38, 39]. Guidelines and open benchmarks for functionally grounded evaluation on tabular data are emerging [40, 41]. Estimators of Shapley values differ in accuracy and cost [42], LIME's neighborhood sampling can be redesigned to raise its fidelity [44], and counterfactual explainers can disagree on the same instance [45]. This study adds a comparison in which the four explainers are paired on the same instances, their configuration is varied, the validity of the stability measurement is tested, and cost is counted in model evaluations as well as in time.

3 Materials and Methods

3.1 Design and Research Questions

All results come from one experimental cohort run for this study under the protocol below (Table 1). An earlier analysis of the SHAP–LIME question used runs collected on several hosts; an audit found that in most of its cells the evaluation instances overlapped the training partition, that one model could not be recovered and that its timings mixed hosts, so no number from it is used here.

For each endpoint Y and each instance, the paired difference is $\Delta_X = Y_{\mathrm{SHAP}} - Y_X$ for $X \in \{\mathrm{LIME}, \mathrm{Anchors}, \mathrm{DiCE}\}$. Three hypotheses state an expected direction; each is tested two-sided, and a difference is read in the stated direction only when its interval excludes zero on that side. Sparsity is reported descriptively:

- **H1 (stability).** SHAP is more stable than X: $\Delta_X(S) > 0$.
- **H2 (cost).** SHAP is more expensive than X: $\Delta_X(C) > 0$, tested in wall-clock time; CPU time and the rows passed to the model are reported alongside, without a test.
- **H3 (faithfulness).** SHAP is more faithful than X: $\Delta_X(F) > 0$ and $\Delta_X(G) > 0$.
- **Sparsity (descriptive).** With its 10-feature budget LIME's sparsity is set by the user, and Anchors returns few features by construction, so a test of sparsity would test a configuration, not a method.

Table 1: Design of the benchmark. The primary benchmark is confirmatory; the other parts are exploratory

Factor	**Levels**
Datasets	Adult (primary, 108 encoded features); Breast Cancer (30), German Credit (61), Diabetes (8), Spambase (57), Phishing (68), Connect-4 (126)
Models	logistic regression, random forest, XGBoost, SVM (RBF), MLP
Explainers	SHAP, LIME (full vector; 10-feature variant), Anchors, DiCE
Seeds	Adult: 5; other datasets: 3
Instances	up to 25 per stratum (TP, TN, FP, FN) of the test split, per model and seed; stability on 3 per stratum
Paired cells	Adult: 25 (model × seed); other datasets: 90
Timing subset	5 instances per stratum, 3 repetitions, one host
Sweep and controls	Section 4.3 and Section 4.4

3.2 Data and Models

Adult [16, 17] is the primary dataset. Six further binary datasets were chosen so that feature dimension crosses feature type: Breast Cancer [27] and German Credit [28] from the UCI repository [26], and Diabetes, Spambase, Phishing and Connect-4 [49, 50, 51, 52] in a 2×2 of numeric or categorical features and lower or higher dimension. For Connect-4, a win is the positive class and the two other outcomes the negative class.

Each dataset has one fixed stratified split (80/20, seed 42), shared by preprocessing, model training and every seed. Numeric features are scaled and categorical features one-hot encoded, both fitted on the training split only. Training is capped at 20,000 rows (a stratified subsample) so that the SVM fits in reasonable time. The five models [18, 19] are trained once per dataset with fixed hyperparameters (supplementary file) and frozen; every

run checks the model file against its recorded hash. A seed therefore changes the evaluation instances and the explainers' randomness, never the model. Evaluation instances are drawn from the test split only, up to 25 from each confusion-matrix stratum; smaller strata are taken whole, and the realized counts are reported. Within a cell, all explainers see the same instances. In small strata, different seeds can draw the same instance; the overlap is measured and reported: the mean Jaccard index of the instance sets of two seeds of the same dataset and model is 0.22 (maximum 0.69).

3.3 Explainers

All explainers receive the same encoded features and explain the predicted probability of the positive class:

- **SHAP:** TreeSHAP for the random forest and XGBoost; KernelSHAP for the other models, with 50 background rows and a budget of 500 coalitions.
- **LIME:** 1,000 samples and the library's default kernel width, $0.75\sqrt{M}$ (≈ 7.79 on Adult). The primary contrast uses the full attribution vector; a variant with the 10-feature budget shows the effect of the budget.
- **Anchors:** precision target 0.95; the attribution vector marks the features of the anchor with 1 and the others with 0.
- **DiCE:** one counterfactual of the opposite class (random search); the attribution of feature j is $|x_j - x'_j|$.

Anchors (a rule) and DiCE (a counterfactual) enter as reference explainers of other kinds: the research questions remain about SHAP and LIME, and the two references show whether a SHAP–LIME difference is specific to the pair or shared by explainers in general. These settings are the libraries' defaults where defaults exist, so that a practitioner's first run is what is measured. KernelSHAP's budget is set explicitly because the default ($2M+2048$ coalitions) made the SVM the cost driver of the whole design; Section 4.3 measures the effect of the default against the explicit budget. Anchors and DiCE return explanations of different kinds, so their vectors measure which features the explanation uses, not signed contributions; the metrics below are comparable across the four methods in that sense.

3.4 Metrics

Let $f(x)$ be the predicted probability of the positive class and $e(x) \in \mathbb{R}^M$ the attribution vector; masking feature j replaces it with its training mean $\bar{x}_j$. The metrics follow common definitions [12, 10, 11]:

1. **Stability** S: the mean cosine similarity over all pairs of $T = 15$ explanations of perturbed copies $x + \varepsilon^{(a)}$, $\varepsilon^{(a)} \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 0.1$, added to every encoded feature (scaled numeric and one-hot columns alike), with

S computed on 3 instances per stratum, Eq. (2):

$$S = \frac{2}{T(T-1)} \sum_{1 \le a < b \le T} \cos\Big(e^{(a)}, e^{(b)}\Big) . \tag{2}$$

The cosine with an all-zero vector is set to 0; such vectors are counted and S is also reported without them. Noise on one-hot columns leaves the data manifold; Section 4.4 repeats S with category-preserving perturbations and without noise.

2. **Fidelity** F: the Pearson correlation, across the M features, between $|e_j(x)|$ and the prediction change $|f(x) - f(\tilde{x}^{(j)})|$ when feature j alone is masked, Eq. (3):

$$F = \rho\Big(\big\{|e_j(x)|\big\}_{j=1}^{M},\ \big\{|f(x) - f(\tilde{x}^{(j)})|\big\}_{j=1}^{M}\Big) . \tag{3}$$

Absolute values are used because the masked change is unsigned. When either vector is constant, F is undefined and set to 0; these cases are counted.

3. **Faithfulness gap** G: the prediction change when the $k = 5$ features with the largest $|e_j(x)|$ are masked together.
4. **Sparsity** P: the share of features with $|e_j(x)| > 10^{-4}$ (lower is sparser).
5. **Cost** C: wall-clock time per explanation on one pinned core, CPU time, and the number of rows the explainer passes to the model (Section 3.6).

3.5 Statistical Analysis

Instances are nested in seeds and seeds in models, so instance-level differences are not independent. For each endpoint and contrast, the instance-level differences Δ_{gsi} of model g, seed s and instance i are fitted by a linear mixed model $\Delta_{gsi} = \mu + u_g + v_{gs} + \epsilon_{gsi}$, with random intercepts for the model and for the seed within the model [30]; μ is the effect, reported with its 95% confidence interval and two-sided Wald p-value; if the model cannot be fitted, a t-test on the cell means is used and the case is reported. Because five models are few clusters, the interval is checked by a cluster bootstrap that resamples models [31]. Only instances valid for both explainers of a contrast enter it. On the cell-mean differences, the standardized paired difference d_z [22], the Hodges–Lehmann estimate [32] and the matched-pairs rank-biserial correlation [33], the effect size of the Wilcoxon signed-rank test [21], are reported. The confirmatory family is four endpoints (S, F, G and wall time) times three contrasts; p-values are Holm-corrected over these twelve tests [23]. The four explainers are ranked on the cell means of S, F and G by the Friedman test with the Nemenyi critical difference [20]. The sweep, the controls and the six further datasets are exploratory and are reported without tests.

3.6 Cost Measurement and Computing Environment

All runs used one laptop (AMD Ryzen 5 4600H, 6 cores, 31.4 GB RAM, Windows 11), Python 3.11 with scikit-learn 1.7.1, XGBoost 3.1.2, SHAP 0.50.0, LIME 0.2.0.1, alibi 0.9.6 (Anchors) and dice-ml 0.12. Quality was computed in parallel worker processes; quality does not depend on speed. Cost was measured in a separate pass on a subset (Table 1): one process with one thread, pinned to one core, on mains power; the explainers interleaved per instance, after one warm-up explanation each; three repetitions, of which the median is used; and measurements paused while any other process used more than 10% of a core or the machine was in use. Cost is tested in wall-clock time, the time elapsed between the start and the end of an explanation, read from the system's high-resolution performance counter: Windows advances a process's CPU time in steps of 15.6 ms, too coarse for explanations that take a few milliseconds, whereas on one pinned core of a quiet host wall time measures the same work at sub-millisecond resolution. CPU time is reported alongside. A wrapper counts every row the explainer passes to the model. Rows evaluated, unlike time, transfer to other hardware: multiplied by the cost of one model evaluation they give the explainer's cost on any host. For logistic regression and the MLP, whose cost per row is known, this is also given in floating-point operations.

4 Results

4.1 Primary Benchmark on Adult

Table 2 reports the effects on the 25 Adult cells. SHAP is the more faithful explainer: its fidelity exceeds that of each other explainer in all 25 cells (Δ_X from 0.38 to 0.65, $d_z \geq 3.3$), and its faithfulness gap exceeds that of LIME and DiCE ($\Delta_X = 0.13$ and 0.31, intervals above zero). Against Anchors the gap is larger in every cell (0.16), but the interval of the mixed model includes zero (-0.01 to 0.33) because the effect varies between models. On stability SHAP is ahead of DiCE (0.34, in 24 of 25 cells); against LIME (0.18, in 19 of 25 cells) and Anchors (0.10, in 19 of 25) the intervals include zero, so H1 is not supported for these two contrasts. The Friedman test ranks SHAP first on all three endpoints (mean rank 1.0 on fidelity and on the faithfulness gap, 1.5 on stability; $p < 10^{-8}$ each) and DiCE last (critical difference 0.94); on stability, Anchors (2.0) ranks ahead of LIME (2.8). All-zero vectors occur only for LIME (0.4% of explanations; 5.8% of its perturbed explanations in the stability measure, 6.3% with the full vector); fidelity is undefined for at most 0.1% of the instances of any explainer. Excluding every instance with an all-zero LIME vector, which leaves 134 of the 300 stability instances, changes no conclusion: against LIME the stability difference stays within noise (0.13, 95% CI -0.07 to 0.32), and the fidelity and faithfulness-gap

differences keep their size (0.38 and 0.13). The distribution of the instance-level differences is shown in the supplementary file.

Table 2: Primary benchmark (Adult, 25 cells). Per-method mean levels and paired effects Δ_X = SHAP $-X$: mixed-model estimate with 95% CI, Holm-adjusted p, standardized paired difference d_z, Hodges–Lehmann estimate (HL) and matched rank-biserial correlation r_{rb}. Cost: wall-clock ms per explanation

Endpoint	X	SHAP	X	Δ_X [95% CI]	p_{Holm}	d_z	HL	r
Stability	LIME	0.74	0.56	0.18 [−0.01, 0.37]	0.25	0.89	0.18	0.7
	Anchors		0.63	0.10 [−0.08, 0.28]	0.31	0.45	0.15	0.4
	DiCE		0.40	0.34 [0.14, 0.54]	0.008	1.63	0.34	0.9
Fidelity	LIME	0.83	0.45	0.38 [0.22, 0.54]	<0.001	3.29	0.40	1.0
	Anchors		0.41	0.42 [0.25, 0.58]	<0.001	11.79	0.42	1.0
	DiCE		0.18	0.65 [0.44, 0.86]	<0.001	5.83	0.67	1.0
Faithf. gap	LIME	0.40	0.26	0.13 [0.05, 0.21]	0.008	1.33	0.12	1.0
	Anchors		0.23	0.16 [−0.01, 0.33]	0.25	2.19	0.16	1.0
	DiCE		0.08	0.31 [0.13, 0.50]	0.008	4.48	0.29	1.0
Cost	LIME	3,925	170	3,756 [561, 6,950]	0.11	0.49	22	0.7
	Anchors		9,283	−5,357 [−7,208, −3,507]	<0.001	−1.27	−4,887	−1.0
	DiCE		1,408	2,517 [−961, 5,995]	0.31	0.34	−1,056	−0.2
Sparsity	all	0.22	LIME 0.98, Anchors 0.06, DiCE 0.02 (descriptive)					

4.2 Cost

Table 3 gives the cost per explanation by model family. Wall time orders the explainers differently by model family. With KernelSHAP, SHAP takes 3 to 25 times LIME's time: 29.6 against 9.4 ms on logistic regression, 52.2 against 11.0 ms on the MLP, and 19.5 against 0.79 s on the SVM, whose kernel evaluations make it the slowest model for every explainer. With TreeSHAP, SHAP costs about as much as LIME on the random forest (23.0 against 22.3 ms) and a ninth of it on XGBoost (1.4 against 13.3 ms). On the six further datasets, which were timed in the same way (exploratory; supplementary file, S9), TreeSHAP stays 5 to 9 times faster than LIME on XGBoost, but on the random forest its cost depends on the dataset: lower than LIME's on Breast Cancer, Diabetes and German Credit, about equal on Spambase, and higher on Phishing and Connect-4 (23.3 against 18.3 and 49.5 against 28.7 ms). Anchors (0.75 to 5.6 s) and DiCE (1.1 to 2.6 s) are slower than SHAP on every model except the SVM. For H2, the mean difference from LIME is driven by the SVM: the mixed-model interval excludes zero (3.8 s, 95% CI 0.56 to 6.95 s), but the Holm-adjusted p is 0.11 and the Hodges–Lehmann difference is 22 ms, so H2 is not supported

against LIME. SHAP is faster than Anchors in all 25 cells, the opposite of H2, and its difference from DiCE is not significant (Table 2). The rows evaluated per explanation, which do not depend on the host, are 1,000 for LIME, 25,001 for KernelSHAP (500 coalitions $\times$ 50 background rows, plus the instance), 2,004 for DiCE, 5,501 (logistic regression) to 23,201 (MLP) for Anchors, and none for TreeSHAP. With the 108 encoded features of Adult, a row costs 218 FLOPs for logistic regression and 22,200 for the MLP, so a KernelSHAP explanation costs 5.5 and 555 MFLOPs, against 0.22 and 22 for LIME. Whether TreeSHAP's cost relative to LIME reverses between XGBoost and the random forest, and whether such a reversal reflects the architecture or only the size of the ensemble, is tested directly: both have 100 trees, and the comparison is repeated with each ensemble at the other's depth (three seeds, 20 instances each; supplementary file, S9). The cost follows the depth of the trees more than the ensemble: lowering the random forest from depth 15 to 6 cuts TreeSHAP's time from 21.7 to 3.2 ms, and raising XGBoost from depth 6 to 15 raises it from 1.4 to 7.4 ms, against 12.5 to 21.6 ms for LIME. At matched depth XGBoost remains 2.3 to 2.9 times cheaper, so the implementation adds a smaller effect. TreeSHAP is thus faster than LIME on both ensembles at depth 6 and on XGBoost at depth 15, and as fast as LIME only on the random forest at depth 15, the setting of the main protocol.

Table 3: Cost per explanation on Adult (one host, one core): median wall time and CPU time, and rows evaluated by the model; theoretical complexity per explanation (T trees, L leaves, D depth; n_s samples or coalitions; n_b background rows). LIME with the full vector; ranges are over the models, the largest always on the SVM; Anchors' rows on XGBoost from a recount (supplementary file, S9). FLOPs for logistic regression and the MLP in the text

Explainer	Model	Complexity	Wall (ms)	CP
SHAP (Tree)	RF, XGB	$O(TLD^2)$, no model calls	23.0, 1.4	15
SHAP (Kernel)	logreg, SVM, MLP	$O(n_s n_b)$ rows	29.6, 19,492, 52.2	31.3, 19,35
LIME	all	$O(n_s)$ rows + fit $O(n_s M^2)$	9.4–786	15
Anchors	all	adaptive (precision target)	749–5,640	742
DiCE	all	iterative search	1,064–2,627	1,063

4.3 Configuration Sensitivity

The sweep varied, on Adult for every model and three seeds, LIME's kernel width and number of samples, and KernelSHAP's budget and background size; KernelSHAP was also run on the two tree ensembles, to separate the method from the TreeSHAP implementation. The grid crosses LIME kernel

widths of 0.75, 1.5, 3, 5, 10 and 15 and the default $0.75\sqrt{M}$ with 250 to 5,000 samples, KernelSHAP budgets of 100 to 1,000 coalitions and the default with 10 to 100 background rows, and TreeSHAP with 10 to 100 background rows; each cell has one instance per stratum, with stability on each.

On Adult, LIME's kernel width decides whether its explanations are usable at all. With widths of 0.75 and 1.5 every explanation is all-zero; with 3 and 5 every perturbed copy (15 of 15, and 8.7 of 15 on average) gives an all-zero vector, so stability is not defined. These widths are left out below. With the default width ($0.75\sqrt{M} \approx 7.8$) about one perturbed vector in 15 is all-zero, and with 10 and 15 none. On the remaining configurations all 12 instances of every model are valid, and the comparison uses them (Figure 1).

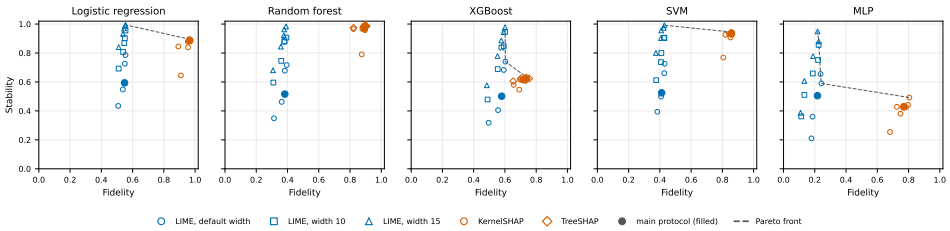


Figure 1: Stability against fidelity over the configuration sweep on Adult, one panel per model: means over the 12 instances valid under every configuration; LIME with the full vector, kernel widths 0.75 to 5 omitted as degenerate. Filled markers: main protocol; dashed line: Pareto front

SHAP's fidelity exceeds LIME's in every pair of configurations on all five models, and its faithfulness gap on four; on XGBoost some LIME configurations with a width of 10 or 15 match it (SHAP ahead in 126 of 180 pairs). The stability ordering, in contrast, depends on LIME's configuration. LIME's stability rises with the kernel width and the number of samples, from 0.34 (default width, 250 samples) to 0.98 (width 15, 5,000 samples), while its fidelity changes little (0.36 to 0.44, mostly with the samples). With a width of 15 and 5,000 samples, LIME is more stable than every SHAP configuration on four models; on the random forest TreeSHAP stays as stable (0.98 to 0.99) and dominates every LIME configuration. On the other models the Pareto front joins SHAP at the fidelity end with wide-kernel LIME at the stability end, so neither explainer dominates.

For KernelSHAP, the library default budget ($2M + 2048 = 2{,}264$ coalitions) and the 500 coalitions of the main protocol give fidelity and stability within 0.01 of each other, except on the MLP (0.81 against 0.77 and 0.49 against 0.43). A budget of 100 coalitions lowers stability by 0.07 to 0.24, and 10 background rows lower fidelity by 0.04 to 0.07. KernelSHAP on the two tree ensembles is within 0.02 of TreeSHAP on both endpoints, so the quality differences between the model families do not come from the TreeSHAP implementation. Cost, timed with the same protocol (supplementary file,

S10), depends on LIME's number of samples and not on its kernel width, and on KernelSHAP's coalitions times background rows. LIME's most stable configuration (width 15, 5,000 samples) takes 4.3 to 5.2 times the default 1,000 samples: 47 against 9.3 ms on logistic regression and 3.9 against 0.77 s on the SVM. It then costs more than SHAP on logistic regression (29 ms) and the tree ensembles (TreeSHAP 1.4 and 25.5 ms), about as much on the MLP, and a fifth as much on the SVM (19.2 s). KernelSHAP's library default costs 3.6 to 4.3 times the 500-coalition budget for almost the same quality, and KernelSHAP on the tree ensembles takes 7 (random forest) and 73 (XGBoost) times TreeSHAP's time: the cost difference, unlike the quality difference, does come from the implementation.

4.4 Validity of the Measurements

Five controls test whether the stability and fidelity measurements measure the explainer: (a) explanations of the unperturbed instance repeated T times, which isolates each explainer's own sampling variance; (b) noise that keeps one-hot features valid (a category is resampled instead of adding Gaussian noise to its indicators); (c) rank correlation and top-k overlap on the full vectors, besides the cosine; (d) the effect of excluding all-zero vectors; and (e) the fidelity ordering under four masking schemes (zero, mean, marginal and grouped replacement, where associated features such as education and its numeric code are masked together) on every model. Controls (a) and (b) were run on Adult for every model and three seeds, on the stability instances; (c) to (e) were computed from the stored vectors of the main runs.

Without input noise (a), LIME's repeated explanations are almost identical (stability 0.98 to 0.99 with ten features and 0.89 to 0.97 with the full vector, 0.77 on the MLP), so its own sampling accounts for 8% of its instability under the main protocol (22% with the full vector); the rest is its response to the noise. Own sampling accounts for about two thirds of the instability of Anchors and for all of that of DiCE. With noise that keeps one-hot features valid (b), LIME with the full vector is more stable than SHAP on all five models and in all 15 cells, and LIME with ten features in 13 of 15, whereas under the Gaussian noise of the main protocol SHAP was ahead: on Adult, the stability ordering of SHAP and LIME depends on whether the perturbed rows can occur. SHAP stays more stable than Anchors and DiCE in all 15 cells under both controls. Rank correlation and top-5 overlap (c) keep SHAP ahead of LIME on four of five models, the MLP being the exception. Excluding all-zero vectors (d) raises LIME's cosine stability by 0.01 to 0.08 and changes no ordering. Under all four masking schemes (e), SHAP's faithfulness gap exceeds that of every other explainer on every Adult model, except against LIME on XGBoost, where the difference is close to zero (0.003 to 0.05). On the six further datasets

SHAP is ahead in 460 of 480 combinations of model, explainer and scheme; most exceptions are Anchors on Connect-4, which is ahead of SHAP on three of five models under every scheme, as under the main protocol.

4.5 Other Datasets

Table 4 reports the four explainers on the six further datasets, now including SHAP stability. SHAP keeps the highest fidelity on all six, ahead of the next explainer by 0.06 (Breast Cancer) to 0.31 (Diabetes), and the largest faithfulness gap on all six (supplementary file). The stability ordering of Adult does not carry over: LIME with the full vector is more stable than SHAP on every further dataset, by 0.05 (Diabetes) to 0.26 (Spambase), and Anchors is also more stable than SHAP on Phishing and Connect-4, the two datasets with categorical features in the 2×2. Neither the number of features nor their type reverses the orderings of SHAP and LIME: both hold in every cell of the 2×2.

Table 4: Other datasets (exploratory): mean over 3 seeds, averaged over the five models; LIME with the full vector. The SD over seeds is below 0.04 in every cell (per cell and per model in the supplementary file)

	Stability				Fidelity			
Dataset	SHAP	LIME	Anch.	DiCE	SHAP	LIME	Anch.	DiCE
Breast Cancer	0.84	0.99	0.38	0.25	0.69	0.50	0.25	0.15
German Credit	0.83	0.94	0.68	0.30	0.77	0.44	0.43	0.18
Diabetes	0.95	1.00	0.82	0.32	0.80	0.49	0.37	0.27
Spambase	0.69	0.95	0.30	0.11	0.64	0.46	0.24	0.13
Phishing	0.75	0.95	0.89	0.25	0.70	0.54	0.51	0.26
Connect-4	0.70	0.85	0.71	0.10	0.80	0.36	0.49	0.09

5 Discussion

5.1 What Holds, and Where

Table 5 states each claim with its evidence and its scope. Confirmatory claims rest on the primary benchmark and its corrected tests; exploratory claims rest on the sweep, the controls and the further datasets, and are reported as patterns, not as tested effects. The main pattern is that SHAP's advantage lies in faithfulness, not in stability: its fidelity is the highest on every model and dataset tested, and its faithfulness gap on every dataset except Connect-4, where Anchors matches it, whereas the stability ordering depends on the dataset and on the noise. LIME is more stable on all six further datasets, and on Adult as well when the noise keeps categories

valid. Cost depends on the SHAP algorithm more than on the explainer: KernelSHAP is the most expensive explainer on the SVM, and TreeSHAP the cheapest on XGBoost. Of the validity controls, only the noise scheme changes an ordering; rank-based stability, the exclusion of all-zero vectors and the masking scheme do not. Over the configuration grid, the fidelity ordering never changes, whereas the stability ordering changes with LIME's kernel width and number of samples: with a width of 15 and 5,000 samples, LIME is the more stable explainer on every model except the random forest.

The recommendations are restricted to what was tested. Where faithfulness matters, SHAP is preferable on every model tested: with XGBoost it costs less than LIME and with the random forest about as much (TreeSHAP), and with the other models it costs 3 to 25 times LIME's time (KernelSHAP). Where stability matters more than faithfulness, LIME with the full vector suffices: it is more stable than SHAP on the six further datasets, and on Adult when the noise keeps categories valid. As feature attributions, Anchors and DiCE are less faithful than SHAP and, except on the SVM, slower. Before relying on a stability comparison, a practitioner should check that the perturbations keep categorical features valid, since Gaussian noise on one-hot indicators reverses the ordering on Adult. LIME's kernel width should be checked next: on Adult, widths up to 5 give all-zero explanations, and the default width gives a stability well below that of a width of 10 or 15 with 2,000 or more samples, at almost the same fidelity. KernelSHAP loses stability below 250 coalitions and fidelity with 10 background rows. The stable LIME setting costs about five times the default, which still leaves it cheaper than KernelSHAP on the SVM but not than TreeSHAP on the tree ensembles.

5.2 Relation to Earlier Work

Earlier comparisons of SHAP and LIME, including a multi-metric benchmark on Adult [9], reported SHAP ahead on fidelity and stability and LIME ahead on cost, for one configuration and with timings from unrecorded hardware. This study differs in four respects: all four explainers are paired on the same instances; every run uses one recorded host and cost is also counted in model evaluations; the configuration is varied and the measurement is tested; and the results extend to six further datasets. The fidelity advantage of SHAP is confirmed. Its stability advantage is qualified: on Adult it is not significant under the original noise and reverses when the noise keeps categories valid, and LIME is the more stable explainer on the six further datasets. The cost statement is qualified as well: LIME is cheaper than KernelSHAP, but not than TreeSHAP on XGBoost. Over the configuration grid, the fidelity statement holds, and the stability statement depends on LIME's kernel width and number of samples as well.

5.3 Limitations

The endpoints compare explainers under a protocol; they do not establish that an explanation is correct, which needs ground truth or human-centered evaluation [29]. Anchors and DiCE vectors encode which features an explanation uses, not signed contributions. Cost was measured on one host; times scale with the hardware, rows evaluated do not. The study covers tabular binary classification with five model families and seven datasets.

6 Conclusions

Whether SHAP outperforms LIME depends on what is asked of the explanation. On the primary benchmark, SHAP is ahead on faithfulness: its fidelity exceeds that of LIME, Anchors and DiCE in every cell of all five model families ($d_z \geq 3.3$), and its faithfulness gap exceeds that of LIME and DiCE; on stability it is ahead only of DiCE (Table 2). The fidelity advantage holds over the whole configuration grid, and the faithfulness-gap advantage on four of five models, whereas the stability ordering depends on LIME's configuration: with a kernel width of 15 and 5,000 samples, LIME is more stable than every SHAP configuration on every model except the random forest (exploratory, Section 4.3). The fidelity advantage also holds on all six further datasets, and the faithfulness-gap advantage on five (Anchors matches SHAP on Connect-4), where LIME is instead the more stable explainer (exploratory, Table 4), as it is on Adult when the stability noise keeps categories valid (Section 4.4). It is paid for in cost only with KernelSHAP, which takes 3 to 25 times LIME's wall time and 25 times its model evaluations (19.5 s per explanation on the SVM), whereas TreeSHAP is faster than LIME on XGBoost and, on the random forest, faster or slower depending on the dataset, and SHAP is faster than Anchors throughout (Table 3). The difference between the ensembles follows tree depth more than the implementation: at matched depth both ensembles cost less with TreeSHAP than with LIME, except the random forest at depth 15. LIME is therefore the cheaper choice for models that need KernelSHAP, such as the SVM, under a tight latency budget, at the price of fidelity. The faithfulness advantage is not specific to the pair: SHAP holds it against Anchors and DiCE as well, and DiCE ranks last on every quality endpoint. The answer is therefore conditional: the choice of explainer depends on the model architecture, the configuration and the latency budget, and a ranking obtained under one configuration should not be transferred to another without measuring it. Whether any of these explainers helps people decide is a separate question that needs a study with participants.

Competing Interests

No competing financial interests or personal relationships that could have appeared to influence the work reported in this paper are declared. No external funding was received for this study.

Data and Code Availability

The code, the frozen models with their hashes, every result row and the analysis scripts are available in a public repository and a versioned archive with a DOI, under the MIT License; their addresses are withheld for masked review and are given on the title page. A supplementary file lists the design, the hyperparameters, the per-dataset and per-model results and the commands that regenerate every table from the result rows.

References

[1] Ribeiro MT, Singh S, Guestrin C. "Why should I trust you?": Explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2016. p. 1135–1144. https://doi.org/10.1145/2939672.2939778

[2] Lundberg SM, Lee SI. A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems 30; 2017. p. 4765–4774.

[3] Retzlaff CO, Angerschmid A, Saranti A, Schneeberger D, Röttger R, Müller H, et al. Post-hoc vs ante-hoc explanations: xAI design guidelines for data scientists. Cognitive Systems Research. 2024;86:101243. https://doi.org/10.1016/j.cogsys.2024.101243

[4] Nauta M, Trienes J, Pathak S, Nguyen E, Peters M, Schmitt Y, et al. From anecdotal evidence to quantitative evaluation methods: A systematic review on evaluating explainability. ACM Computing Surveys. 2023;55(13s):1–42. https://doi.org/10.1145/3583558

[5] Canha D, Kubler S, Främling K, Fagherazzi G. A functionally-grounded benchmark framework for XAI methods: Insights and foundations from a systematic literature review. ACM Computing Surveys. 2025;57(12):1–40. https://doi.org/10.1145/3737445

[6] Longo L, Brcic M, Cabitza F, Choi J, Confalonieri R, Del Ser J, et al. Explainable artificial intelligence (XAI) 2.0: A manifesto of open

challenges and interdisciplinary research directions. Information Fusion. 2024;106:102301. https://doi.org/10.1016/j.inffus.2024.102301

[7] Pawlicki M, Pawlicka A, Uccello F, Szelest S, D'Antonio S, Kozik R, et al. Evaluating the necessity of the multiple metrics for assessing explainable AI: A critical examination. Neurocomputing. 2024;602:128282. https://doi.org/10.1016/j.neucom.2024.128282

[8] Zheng X, Shirani F, Chen Z, Lin C, Cheng W, Guo W, et al. F-Fidelity: A robust framework for faithfulness evaluation of explainable AI. In: The 13th International Conference on Learning Representations (ICLR); 2025. https://openreview.net/forum?id=X0r4BN50Dv

[9] Herrera-Vásquez J, Herrero-Uceda M. A framework for rigorous evaluation of model-agnostic explainability methods: multi-metric statistical benchmarking, operational protocol, and reproducibility. Revista de Investigación Multidisciplinaria Iberoamericana (RIMI). 2026;(3). https://doi.org/10.69850/rimi.vi3.307

[10] Bhatt U, Weller A, Moura JMF. Evaluating and aggregating feature-based model explanations. In: Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI); 2020. p. 3016–3022. https://doi.org/10.24963/ijcai.2020/417

[11] Samek W, Binder A, Montavon G, Lapuschkin S, Müller KR. Evaluating the visualization of what a deep neural network has learned. IEEE Transactions on Neural Networks and Learning Systems. 2017;28(11):2660–2673. https://doi.org/10.1109/TNNLS.2016.2599820

[12] Alvarez-Melis D, Jaakkola TS. On the robustness of interpretability methods. In: ICML Workshop on Human Interpretability in Machine Learning (WHI 2018); 2018. https://arxiv.org/abs/1806.08049

[13] Hedstrom A, Weber L, Bareeva D, Krakowczyk D, Motzkus F, Samek W, et al. Quantus: An explainable AI toolkit for responsible evaluation of neural network explanations and beyond. Journal of Machine Learning Research. 2023;24:1–11. https://jmlr.org/papers/v24/22-0142.html

[14] Agarwal C, Ley D, Krishna S, Saxena E, Pawelczyk M, Johnson N, et al. OpenXAI: Towards a transparent evaluation of model explanations. In: 36th Conference on Neural Information Processing Systems, Datasets and Benchmarks Track; 2022. https://openreview.net/forum?id=MU2495w47rz

[15] Adebayo J, Gilmer J, Muelly M, Goodfellow I, Hardt M, Kim B. Sanity checks for saliency maps. In: Advances in Neural Information Processing Systems 31; 2018. p. 9525–9536.

[16] Kohavi R. Scaling up the accuracy of Naive-Bayes classifiers: A decision-tree hybrid. In: Proceedings of the 2nd International Conference on Knowledge Discovery and Data Mining; 1996. p. 202–207.

[17] Becker B, Kohavi R. Adult [dataset]. UCI Machine Learning Repository; 1996. https://doi.org/10.24432/C5XW20

[18] Breiman L. Random forests. Machine Learning. 2001;45(1):5–32. https://doi.org/10.1023/A:1010933404324

[19] Chen T, Guestrin C. XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining; 2016. p. 785–794. https://doi.org/10.1145/2939672.2939785

[20] Demšar J. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research. 2006;7:1–30. https://jmlr.org/papers/v7/demsar06a.html

[21] Wilcoxon F. Individual comparisons by ranking methods. Biometrics Bulletin. 1945;1(6):80–83. https://doi.org/10.2307/3001968

[22] Lakens D. Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. Frontiers in Psychology. 2013;4:863. https://doi.org/10.3389/fpsyg.2013.00863

[23] Holm S. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics. 1979;6(2):65–70.

[24] Ribeiro MT, Singh S, Guestrin C. Anchors: High-precision model-agnostic explanations. In: Proceedings of the 32nd AAAI Conference on Artificial Intelligence; 2018. p. 1527–1535. https://doi.org/10.1609/aaai.v32i1.11491

[25] Mothilal RK, Sharma A, Tan C. Explaining machine learning classifiers through diverse counterfactual explanations. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency; 2020. p. 607–617. https://doi.org/10.1145/3351095.3372850

[26] Dua D, Graff C. UCI Machine Learning Repository. Irvine (CA): University of California, School of Information and Computer Science; 2019. http://archive.ics.uci.edu/ml

[27] Wolberg W, Mangasarian O, Street N, Street W. Breast cancer Wisconsin (diagnostic) [dataset]. UCI Machine Learning Repository; 1993. https://doi.org/10.24432/C5DW2B

[28] Hofmann H. Statlog (German credit data) [dataset]. UCI Machine Learning Repository; 1994. https://doi.org/10.24432/C5NC77

[29] Haufe S, Wilming R, Clark B, Zhumagambetov R, Boubekki A, Martin J, et al. Explainable AI needs formalization. npj Artificial Intelligence. 2026;2:42. https://doi.org/10.1038/s44387-026-00095-1

[30] Bates D, Mächler M, Bolker B, Walker S. Fitting linear mixed-effects models using lme4. Journal of Statistical Software. 2015;67(1):1–48. https://doi.org/10.18637/jss.v067.i01

[31] Field CA, Welsh AH. Bootstrapping clustered data. Journal of the Royal Statistical Society: Series B. 2007;69(3):369–390. https://doi.org/10.1111/j.1467-9868.2007.00593.x

[32] Hodges JL, Lehmann EL. Estimates of location based on rank tests. The Annals of Mathematical Statistics. 1963;34(2):598–611. https://doi.org/10.1214/aoms/1177704172

[33] Kerby DS. The simple difference formula: An approach to teaching nonparametric correlation. Comprehensive Psychology. 2014;3:11.IT.3.1. https://doi.org/10.2466/11.IT.3.1

[34] Hooshyar D, Yang Y. Problems with SHAP and LIME in interpretable AI for education: A comparative study of post-hoc explanations and neural-symbolic rule extraction. IEEE Access. 2024;12:137472–137490. https://doi.org/10.1109/ACCESS.2024.3463948

[35] Schulte L, Ledel B, Herbold S. Studying the explanations for the automated prediction of bug and non-bug issues using LIME and SHAP. Empirical Software Engineering. 2024;29(4):93. https://doi.org/10.1007/s10664-024-10469-1

[36] Lazar A, Pokhrel P, Das S. Beyond accuracy: A comprehensive comparative study of gradient boosting versus tabular deep learning and explainability techniques for mixed-type tabular data models using SHAP and LIME. International Journal on Artificial Intelligence Tools. 2026;35(2):2640003. https://doi.org/10.1142/S0218213026400038

[37] Ababneh M, Aljarrah A. Faithfulness and stability of black box explanations for tabular and image classifiers. Big Data and Cognitive Computing. 2026;10(11):362. https://doi.org/10.3390/bdcc10110362

[38] Salih AM, Raisi-Estabragh Z, Galazzo IB, Radeva P, Petersen SE, Lekadir K, et al. A perspective on explainable artificial intelligence methods: SHAP and LIME. Advanced Intelligent Systems. 2025;7(1):2400304. https://doi.org/10.1002/aisy.202400304

[39] Mersha M, Lam K, Wood J, AlShami AK, Kalita J. Explainable artificial intelligence: A survey of needs, techniques, applications, and future direction. Neurocomputing. 2024;599:128111. https://doi.org/10.1016/j.neucom.2024.128111

[40] Velmurugan M, Ouyang C, Xu Y, Sindhgatta R, Wickramanayake B, Moreira C. Developing guidelines for functionally-grounded evaluation of explainable artificial intelligence using tabular data. Engineering Applications of Artificial Intelligence. 2025;141:109772. https://doi.org/10.1016/j.engappai.2024.109772

[41] Moiseev I, Balabaeva K, Kovalchuk S. Open and extensible benchmark for explainable artificial intelligence methods. Algorithms. 2025;18(2):85. https://doi.org/10.3390/a18020085

[42] Olsen LHB, Glad IK, Jullum M, Aas K. A comparative study of methods for estimating model-agnostic Shapley value explanations. Data Mining and Knowledge Discovery. 2024;38(4):1782–1829. https://doi.org/10.1007/s10618-024-01016-z

[43] Kapoor S, Cantrell EM, Peng K, et al. REFORMS: Consensus-based recommendations for machine-learning-based science. Science Advances. 2024;10(18):eadk3452. https://doi.org/10.1126/sciadv.adk3452

[44] Saadatfar H, Kiani-Zadegan Z, Ghahremani-Nezhad B. US-LIME: Increasing fidelity in LIME using uncertainty sampling on tabular data. Neurocomputing. 2024;597:127969. https://doi.org/10.1016/j.neucom.2024.127969

[45] Brughmans D, Melis L, Martens D. Disagreement amongst counterfactual explanations: How transparency can be misleading. TOP. 2024;32(3):429–462. https://doi.org/10.1007/s11750-024-00670-2

[46] Baniecki H, Biecek P. Adversarial attacks and defenses in explainable artificial intelligence: A survey. Information Fusion. 2024;107:102303. https://doi.org/10.1016/j.inffus.2024.102303

[47] Huang X, Marques-Silva J. On the failings of Shapley values for explainability. International Journal of Approximate Reasoning. 2024;171:109112. https://doi.org/10.1016/j.ijar.2023.109112

[48] Bilodeau B, Jaques N, Koh PW, Kim B. Impossibility theorems for feature attribution. Proceedings of the National Academy of Sciences. 2024;121(2):e2304406120. https://doi.org/10.1073/pnas.2304406120

[49] Smith JW, Everhart JE, Dickson WC, Knowler WC, Johannes RS. Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. In: Proceedings of the Annual Symposium on Computer Application in Medical Care; 1988. p. 261–265.

[50] Hopkins M, Reeber E. Spambase [dataset]. UCI Machine Learning Repository; 1999. https://doi.org/10.24432/C53G6X

[51] Mohammad R, McCluskey L. Phishing websites [dataset]. UCI Machine Learning Repository; 2012. https://doi.org/10.24432/C51W2X

[52] Tromp J. Connect-4 [dataset]. UCI Machine Learning Repository; 1995. https://doi.org/10.24432/C59P43

Table 5: What holds, where, and under which configuration

Claim	**Status**	**Scope**
SHAP more stable than X	confirmatory	Against DiCE only, in all five model families of Adult; against LIME and Anchors not significant
SHAP more faithful than X	confirmatory	Fidelity against all three in every Adult cell; faithfulness gap against LIME and DiCE, not Anchors
SHAP more expensive than X	confirmatory	Not supported: against LIME only with KernelSHAP (3–25 times), not significant after Holm; SHAP faster than Anchors in every cell; DiCE not significant
Orderings over the configuration grid	exploratory	Fidelity: SHAP ahead in every pair of configurations on all five models; faithfulness gap on four (not XGBoost). Stability: depends on LIME's width and samples; LIME (width 15, 5,000 samples) ahead except on the random forest. Widths 0.75 to 5 degenerate
Orderings under the validity controls	exploratory	Stability: LIME ahead of SHAP on Adult with valid one-hot noise; other controls change no ordering. Faithfulness gap: same ordering under all four masking schemes
Orderings on other datasets	exploratory	SHAP highest on fidelity on all six and on faithfulness gap on five (Anchors equal on Connect-4); LIME (full vector) more stable on all six, Anchors on Phishing and Connect-4
TreeSHAP vs LIME cost by ensemble	exploratory	XGBoost: TreeSHAP faster on all seven datasets. Random forest: faster on three, equal on two, slower on two. Cost follows tree depth: at matched depth XGBoost 2.3–2.9 times cheaper